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Total No. of printed pages = 02

Monsoon, 2024
B.Tech 1st Semester Examination
Physics

Course Code: BPY23111T

Full Marks – 50

Time –2 hours

The figure in the margin indicates full marks for the questions.

Section 1

1. Fill up the blanks: **Marks=10**

- (a) The del operator contains partial differentiations components along X, Y and _____ axes.
- (b) The gradient of a scalar function gives _____ quantity.
- (c) The concept of displacement current comes from the inconsistency of _____ law.
- (d) If L is the inductance, C the capacitance and R, the resistance, I, the current at time, t, then, the value of the induced emf is _____.
- (e) From Maxwell's equations, $\vec{\nabla} \cdot \vec{B} =$ _____
- (f) By the use of stop, _____ rays can be blocked in spherical aberration.
- (g) Colour defect in lenses is called _____ aberration.
- (h) In Newton's rings, in the transmitted system, the central fringe is always _____.
- (i) If angle is 2π and angular velocity is ω , then the time period, T is _____
- (j) In free vibration, the object vibrates by its own characteristic frequency, called _____ frequency.

Section 2

2. Answer any **one** question **Marks=10**

- (a) If $\vec{A} = \hat{i}(5x^2+2z) - 3y^3\hat{j} + (4z+2x)\hat{k}$, find, $\vec{\nabla} \cdot \vec{A}$. Write the Maxwell's equations and explain the physical significance of each equation. (5+5=10)
- (b) Write the Biot Savart law. What is dielectric polarisation? What is a metastable state? Derive the equation of Simple Harmonic Motion. (2+2+2+4=10)

Section 3

3. Answer any **one** question. **Marks=10**

- (a) Explain the inconsistency of Ampere's law. Find the polarisation of dielectric sphere in uniform field. (5+5=10)
- (b) What are Einstein's coefficients? Explain the working principle of He-Ne laser. (5+5=10)

Section 4

4. Answer any **one** question. **Marks=10**

- (a) Explain with a neat diagram, the method of production of Newton's rings. Newton's rings are observed between plano convex lens of radius of curvature, 120 cm and a plane glass plate. The diameter of the 15th bright ring is 0.58 cm. Calculate, (i) the diameter of the 25th bright ring and (ii) the wavelength of the light used. (5+5=10)
- (b) Derive Poisson's and Laplace equations. 10

Section 5

5. Answer any **one** question. **Marks=10**

- (a) State Faraday's laws of electromagnetic induction. Write Lenz's law. Apply it to find the value of motional emf. (4+2+4=10)
- (b) Prove that the measure of chromatic aberration is equal to the product of the dispersive power and focal length. For an achromatic doublet, find the condition of achromatism. (5+5=10)